

Lecture 15: Convergence in Probability, Central Limit Theorem

Jamie Haddock

Table of contents

- 1 Convergence in probability: `{.callout-note icon=false}## Definition: Convergence in probability`
 $Y_n \xrightarrow{p} c$ if for every $\varepsilon > 0$,

$$P(|Y_n - c| \geq \varepsilon) \rightarrow 0.$$

The WLLN is exactly the statement $\bar{X}_n \xrightarrow{p} \mu$: `{.callout-warning icon=false}## Theorem: Central Limit Theorem`
For iid $X_1, \dots, X_n$ with mean μ , variance σ^2 :

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1).$$

For large n , $\bar{X}_n \approx N(\mu, \sigma^2/n)$ – regardless of the distribution of the X_i : `## Watching the CLT happen`

1.1 Reading & Homework *Reading: §5.3-5.4 (Bertsekas & Tsitsiklis, Introduction to Probability).* 2

—title: “Lecture 15: Convergence in Probability, Central Limit Theorem”author: “Jamie Haddock”format: revealjs: output-file: Lecture15_slides slide-number: true preview-links: auto logo: figs/hmc.png css: input/slides.css incremental: true smaller: true code-fold: true embed-resources: true html: code-fold: true embed-resources: true pdf: documentclass: article toc: true number-sections: true geometry: - top=lin - left=lin - bottom=lin - right=linformat-links: falsejupyter: python3filters: - input/remove-pause.luaexecute: echo: true eval: true—

- 1 Convergence in probability: `{.callout-note icon=false}## Definition: Convergence in probability`
 $Y_n \xrightarrow{p} c$ if for every $\varepsilon > 0$,

$$P(|Y_n - c| \geq \varepsilon) \rightarrow 0.$$

The WLLN is exactly the statement $\bar{X}_n \xrightarrow{p} \mu$: `{.callout-warning icon=false}## Theorem: Central Limit Theorem`
For iid $X_1, \dots, X_n$ with mean μ , variance σ^2 :

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1).$$

For large n , $\bar{X}_n \approx N(\mu, \sigma^2/n)$ – regardless of the distribution of the X_i : `## Watching the CLT happen`

```

from scipy.stats import normfig, axes = plt.subplots(1, 3, figsize=(12, 3.5))for ax, n in zip(axes, [1,
—# Case Study: Kaczmarz Iterates Converge in ProbabilityRandomized Kaczmarz satisfies  $E[\|x_k - x^*\|^2] \leq$ 
 $\rho^k \|x_0 - x^*\|^2$  for some  $\rho < 1$ . Combined with Markov's inequality (L14), this implies  $x_k \xrightarrow{p} x^*$ .
m, n = 40, 10A = rng.normal(size=(m, n))x_true = rng.normal(size=n)b = A @ x_truep = np.sum(A**2, axis=

```

1.1 Reading & Homework

Reading: §5.3-5.4 (Bertsekas & Tsitsiklis, Introduction to Probability).